

CASTLEPOINT, New Zealand

WMO: 934980

Lat: 40.9042S

Lon: 176.2119E

Elev: 394

StdP: 14.49

Time Zone: 12.00 (NZL)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	40.0	41.5	31.0	25.6	43.9	32.9	27.8	44.4	70.1	50.0	59.4	52.8	17.0	240	0.839

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.7	76.3	64.0	73.6	62.8	71.4	61.9	66.1	72.7	64.7	70.4	63.5	68.8	22.7	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.9	90.1	68.1	62.5	86.0	66.6	61.4	82.4	65.8	31.0	72.8	30.0	70.6	29.0	68.8	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
56.1	49.5	43.6	37.4	81.8	1.3	3.2	36.4	84.1	35.7	86.0	34.9	87.8	33.9	90.1		
			34.6	69.2	1.3	1.7	33.7	70.4	32.9	71.5	32.2	72.4	31.3	73.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.1	63.1	63.6	61.6	57.7	54.6	50.9	49.5	49.7	52.1	54.3	56.5	60.2
	DBStd	6.53	4.88	4.66	4.42	4.23	4.50	3.96	3.65	3.70	4.21	4.26	4.67	4.86
	HDD50	195	0	0	0	2	12	35	54	50	26	12	4	0
	HDD65	3355	95	76	124	221	322	422	480	474	388	333	257	163
	CDD50	2422	406	380	359	232	155	64	39	40	88	144	199	316
	CDD65	108	36	36	18	2	0	0	0	0	0	0	2	14
	CDH74	208	80	86	19	0	0	0	0	0	0	0	2	21
	CDH80	13	6	7	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	19.3	19.8	17.9	17.6	16.8	18.4	18.6	19.4	18.8	20.7	22.0	21.5	19.7
(15) Precipitation	PrecAvg	42.8	2.7	3.0	3.1	3.2	3.8	4.5	5.6	4.3	3.2	3.8	2.9	2.6
	PrecMax	60.1	9.7	11.0	9.6	11.6	8.7	9.2	11.2	9.1	8.1	8.4	6.7	5.9
	PrecMin	24.2	0.4	0.7	0.7	0.5	0.5	2.2	2.1	1.7	1.0	1.8	0.7	0.6
	PrecStd	8.6	2.0	2.4	1.9	2.7	1.8	1.6	2.4	1.6	1.6	1.7	1.5	1.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.0	80.9	76.4	70.9	66.9	62.3	60.5	60.8	64.7	67.7	72.4	76.6
		MCWB	65.1	66.7	62.4	61.3	59.5	56.1	54.4	53.7	56.2	57.7	60.8	63.6
	2%	DB	76.0	76.6	72.8	67.6	64.8	60.1	58.0	58.3	62.2	64.8	68.4	73.2
		MCWB	63.9	64.7	62.0	59.0	58.0	53.9	52.1	51.7	54.7	56.3	58.9	62.1
	5%	DB	72.9	73.5	70.2	65.6	62.6	58.4	56.2	56.5	60.2	62.9	65.9	70.3
		MCWB	62.3	63.6	60.9	58.3	56.0	52.8	50.8	50.4	53.3	55.4	57.8	61.0
	10%	DB	70.4	70.8	67.9	63.6	60.8	56.9	54.7	54.9	58.4	61.0	63.7	67.7
		MCWB	61.6	62.4	60.3	57.6	55.1	52.1	49.9	49.6	52.1	54.1	56.2	59.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.2	68.7	65.2	63.5	61.0	57.6	55.5	54.8	57.9	59.6	62.6	66.1
		MCDB	74.8	76.0	71.0	67.1	64.9	60.5	58.7	58.7	62.4	65.3	69.2	72.8
	2%	WB	66.1	66.3	63.6	61.4	59.1	55.4	53.2	53.0	55.7	57.6	60.5	63.9
		MCDB	72.2	73.5	69.1	64.9	62.5	58.1	56.4	56.2	60.6	63.0	66.3	69.5
	5%	WB	64.5	64.9	62.6	60.1	57.7	53.9	51.8	51.6	54.2	56.2	58.8	62.4
		MCDB	69.8	71.0	67.3	63.6	60.9	57.3	55.1	54.8	58.7	61.3	63.8	67.7
	10%	WB	63.1	63.6	61.6	58.8	56.3	52.6	50.5	50.5	52.9	55.0	57.4	61.0
		MCDB	68.3	68.9	66.2	62.5	59.9	56.4	53.8	54.0	57.1	59.8	62.1	66.0
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	9.7	8.9	8.2	7.4	7.1	6.4	7.0	8.2	9.4	9.5	9.5
		MCDBR	14.3	14.2	12.5	10.7	8.8	8.4	7.6	9.0	10.3	11.7	12.8	13.4
		MCWBR	7.7	6.9	6.4	6.4	6.0	6.2	5.5	6.3	7.0	7.5	7.9	7.4
	5% WB	MCDBR	11.6	11.7	9.7	7.9	7.1	7.2	6.6	7.2	9.0	10.5	11.0	11.1
		MCWBR	7.3	7.2	6.2	6.2	5.9	6.2	5.6	5.9	6.8	7.6	7.6	7.4
(40) Clear-Sky Solar Irradiance	taub	0.338	0.331	0.322	0.307	0.290	0.294	0.289	0.298	0.317	0.328	0.322	0.334	(40)
	taud	2.558	2.597	2.609	2.645	2.670	2.618	2.640	2.627	2.549	2.532	2.575	2.579	(41)
	Ebn at Noon	314	308	298	282	267	253	264	281	294	306	318	317	(42)
	Edh at Noon	33	31	28	23	19	18	19	23	29	32	33	33	(43)
(44) All-Sky Solar Radiation	RadAvg	2091	1814	1408	961	650	484	557	808	1190	1597	1936	2037	(44)
	RadStd	123	164	91	71	42	33	43	56	83	95	149	142	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	+0.58	+0.90	N/A	N/A	N/A	N/A	-45	N/A	N/A	N/A

Nomenclature: See separate page